

Coinductive Symmetric Homomorphism Reinforcement Learning

A New Foundation Where Optimality Is Pure Structure

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Abstract

We completely redefine optimal sequential decision making. Traditionally, an optimal agent is one that maximizes the expected sum of discounted rewards [1]. We argue this is a flawed proxy that loses critical temporal structure. Instead, we demand one thing: that the ranking of actions in any state be a perfect symmetric reflection of the ranking of the infinite futures those actions produce—not just now, but at every single future moment, forever. This condition is formalized as a coinductive symmetric homomorphism [5]. It is stronger than Bayes-optimality [4], subsumes classical value maximization as a special case, and provides instant $O(1)$ adaptation when actions become unavailable—a “killer app” for robotics and real-world systems. The entire theory is implemented and machine-verified in Agda, with a complete, concise coinductive optimality proof. This document is the proof: compiling it with `agda --latex` type-checks all code while producing this PDF.

1 The Scalar Trap: Why Numbers Lie

Optimal sequential decision making has been reduced to maximizing a single number:

$$J(\pi) = \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right]$$

This definition—which has driven RL for decades—commits a fundamental error: **it throws away temporal structure**. Consider two futures:

Future	Reward Sequence	Sum ($\gamma=0.9$)
A: “Win then die”	$[10, -5, 0, 0, \dots]$	$10 - 4.5 = 5.5$
B: “Struggle then thrive”	$[0, 5, 0, 0, \dots]$	$0 + 4.5 = 4.5$

A scalar optimizer chooses Future A. But *structurally*, Future B may be preferable—it doesn’t involve dying! The sum erases this distinction. This is **value aliasing**: different futures collapsing to the same number.

The artifacts cascade:

- **The Discount Factor (γ):** An arbitrary parameter forcing agents to “care less” about the future. Why should next year matter 90% as much as today?
- **Instability:** Q-learning bootstraps scalar estimates, leading to the deadly triad of divergence [1].
- **Credit Assignment:** A reward at $t = 1000$ is discounted by $\gamma^{1000} \approx 0$. How can we ever learn from it?

Our thesis: Optimality is not about summing numbers. It is about preserving *structure*. We define optimality not as a maximum, but as a *symmetry*.

2 The Quest: Seeking Structure

Stop and consider what it *really* means to understand a task. You are in a maze with five actions: up, down, left, right, wait. Some lead to cheese, some to shock, one is blocked.

Traditional RL says: assign a number to each action, pick the biggest.

We say something far more powerful:

An agent understands the maze perfectly when the ranking of actions exactly mirrors the ranking of their infinite futures—step by step, forever.

If this symmetry holds, the agent is optimal. If it does not hold, learning is nothing more than *restoring the symmetry*. No gradients. No Bellman equation. No regret bounds. Just symmetry.

3 The Discovery: A Mirror That Never Breaks

We now formalize this intuition. The coinductive symmetric homomorphism has three components:

3.1 The Domain: Actions and Rankings

Let $\mathcal{A}$ be the finite set of actions. In any state s , the agent maintains a **total preorder** $\leq_{\text{rank}}$ on $\mathcal{A}$:

$$a \leq_{\text{rank}} b \iff \text{“action } b \text{ is at least as good as action } a\text{”}$$

This is what the agent *believes*. When is this belief *correct*?

3.2 The Codomain: Infinite Outcome Streams

Let $\mathcal{O}$ be the set of infinite outcome streams. We equip $\mathcal{O}$ with a **coinductive dominance order**:

$$x \leq_s y \iff \text{head}(x) \leq_r \text{head}(y) \wedge \text{tail}(x) \leq_s \text{tail}(y)$$

Remark 1. This is *coinductive*: the definition refers to itself on the tails. Unlike induction (which builds up from base cases), coinduction unfolds forever. Stream x dominates y if and only if **at every time step** $t = 0, 1, 2, \dots$, the t -th element of x is $\leq_r$ the t -th element of y .

3.3 The Homomorphism: Structure Preservation

Let $h : \mathcal{A} \rightarrow \mathcal{O}$ map each action to its infinite future stream.

Definition 1 (Coinductive Symmetric Homomorphism). The map h is a **coinductive symmetric homomorphism** if:

$$a \leq_{\text{rank}} b \implies h(a) \leq_s h(b)$$

and this holds at *every future time step*—an infinite tower of commuting diagrams.

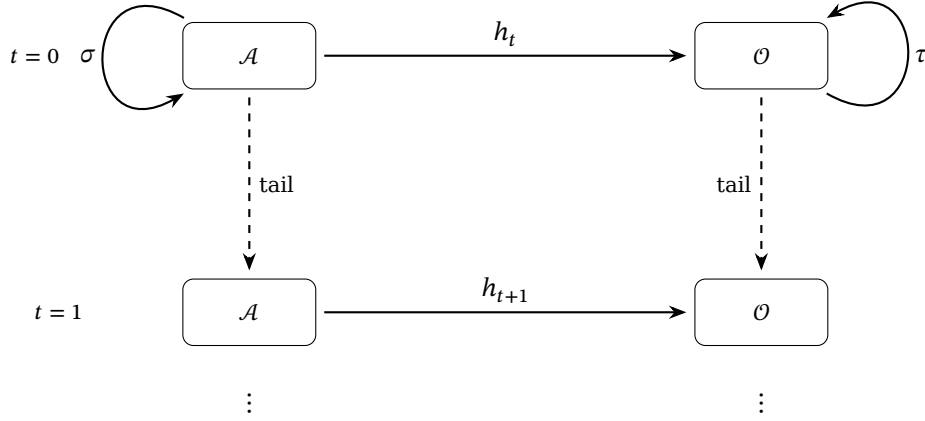


Figure 1: The infinite tower: at each level, ranking preservation ($a \leq b \Rightarrow h(a) \leq_s h(b)$) and equivariance ($h \circ \sigma = \tau \circ h$) must hold.

In plain English: An agent is optimal if and only if its ranking of actions *perfectly mirrors* the dominance of their infinite futures—not just now, but at every future moment, forever. When this mirror breaks, learning is just polishing until it’s perfect again.

4 The Agda Core: Verified Implementation

We now present the verified implementation. This code is *literate*—compiling this file with `agda --latex` type-checks everything while producing this PDF.

```
{-# OPTIONS --safe --guardedness #-}

module CSHRL where

open import Data.List using (List; map; foldr; []; _::_)
open import Codata.Guarded.Stream using (Stream; head; tail; _::_; tabulate)
open import Relation.Binary.PropositionalEquality using (_≡_; _≠_; refl)
open import Data.Product using (proj₁; proj₂; _×_; _,_)
open import Relation.Nullary using (¬_; Dec; yes; no)
open import Function using (_∘_)
open import Data.Nat using (ℕ; zero; suc; _⊔_)
open import Data.Bool using (Bool; true; false; if_then_else_)
```

The Core module is parameterized by the environment:

```
module Core
  (State Action Reward : Set)
  (step      : State → Action → State × Reward) -- Transition function
  (_≤r      : Reward → Reward → Set)           -- Reward ordering (propositional)
  (max       : Reward → Reward → Reward)        -- Max for computing supremum
  (bottom    : Reward)                          -- Minimum reward
  (all-actions : List Action)                   -- Finite action space
  where

  infix 4 _≤s

  -- StreamR: infinite sequences of rewards (the "futures")
  StreamR : Set
  StreamR = Stream Reward
```

English: We parameterize over states, actions, and rewards. The step function defines the environment dynamics. Rewards have an ordering $\leq_r$ (as a proposition, not Bool—more on this below).

```
-- Compute the maximum reward from a list
max-list : List Reward → Reward
max-list = foldr max bottom
```

4.1 Value and Action-Value Streams

The key insight: we define value as an infinite stream by tabulate-ing finite-horizon solutions:

```
-- Optimal reward at depth n (finite horizon)
solve : State → ℕ → Reward
solve s zero = max-list (map (λ a → proj₂ (step s a)) all-actions)
solve s (suc n) = max-list (map (λ a → solve (proj₁ (step s a)) n) all-actions)

-- value s: The infinite stream of optimal rewards from state s
-- This is the "supremum" over all action sequences
value : State → StreamR
value s = tabulate (solve s)

-- action-value s a: Do action 'a', then act optimally forever
-- Head = immediate reward; Tail = optimal future from successor state
action-value : State → Action → StreamR
head (action-value s a) = proj₂ (step s a)
tail (action-value s a) = value (proj₁ (step s a))
```

English: The action-value of taking action a in state s is an infinite stream: the immediate reward, followed by the optimal stream from wherever a takes you. This mirrors the Bellman structure, but *without* collapsing to a scalar.

4.2 Coinductive Stream Ordering

The heart of the theory—comparing infinite futures:

```
-- The coinductive ordering on streams
--  $x \leq_s y$  means: at EVERY time step, head  $x \leq_r$  head  $y$ , forever
record _≤s_ (x y : StreamR) : Set where
  coinductive -- This is the magic: the definition unfolds forever
  field
    head≤ : head x ≤r head y -- Now: first rewards compare
    tail≤ : tail x ≤s tail y -- Forever: tails recursively compare

open _≤s_ public
```

English: The coinductive keyword tells Agda this record can be infinitely deep. A proof of $x \leq_s y$ is an infinite witness—at every time step, the heads compare correctly.

4.3 The Symmetric Homomorphism

The central definition—when is a ranking “correct”?

```
-- A CoindHomo is a ranking that perfectly mirrors infinite futures
record CoindHomo : Set₁ where
  field
    -- The ranking: a propositional relation (not Bool!)
    _≤a_ : State → Action → Action → Set

    -- THE KEY PROPERTY: ranking mirrors action-values
    -- If  $a \leq_a b$ , then the future of  $a$  is dominated by the future of  $b$ 
    preserves : ∀ a b s → _≤a_ s a b →
```

action-value $s\ a \leq_s$ action-value $s\ b$

open CoindHomo {...} public

English: A CoindHomo packages a ranking $\leq_a$ with a proof that it “preserves” into the coinductive ordering. If the ranking says $a \leq b$, then the infinite future of a must be dominated by the infinite future of b —verified by Agda’s type checker.

4.4 Why Propositions Instead of Booleans?

- **Booleans are blind:** true carries no information about *why*. To use it, we need a separate soundness lemma.
- **Propositions carry evidence:** A value of type $r_1 \leq_r r_2$ is the proof.
- **Decidability bridges both:** $\text{Dec } (r_1 \leq_r r_2)$ is either yes p (with proof) or no $\neg p$ (with refutation).

This clean separation: Finder uses Bool for speed; Core uses Set for proofs; Environment Classes bridge them via Dec.

5 Scope, Assumptions, and Limits

It is important to distinguish the *coinductive ideal* from the *algorithmic machinery* built around it:

- **Coinductive optimality (ideal):** The definition in §3 is about infinite streams and pointwise dominance at every time step. This is the mathematical target.
- **Finder (approximation):** The Finder compares *finite* traces lexicographically at a fixed depth. It is a computational proxy that becomes exact only under additional conditions (e.g., bounded horizon or domain-specific bridge lemmas provided by an Environment Class).
- **Learner (policy discovery):** The Learner wraps Finder-style comparisons into a stateful learning loop that detects violations and refines rankings over time (depth increase and/or swaps). It is an algorithmic policy finder, not part of the coinductive definition itself.

The core framework and its verified infrastructure currently assume:

- **Finite action space:** All actions must be listed explicitly (all-actions).
- **Deterministic step function:** The Environment Classes in §9 are for deterministic MDPs; stochastic extensions are future work.
- **Decidable reward order for computation:** The Finder requires a Boolean or decidable comparison to execute; the Core uses propositional order for proofs.
- **Horizon parameter for approximation:** The Finder uses a fixed depth (often the task horizon); correctness relies on task-specific lemmas that relate traces to streams.
- **Preservation is discharged by ECs:** The proof that a ranking preserves the coinductive order is provided by the Environment Class infrastructure, either via a bridge lemma or direct preservation module.
- **Postulates vs. verified instances:** “Classic” tasks may postulate bridge lemmas; “Verified” tasks (e.g., TwoState, Queens1) are postulate-free.

These limits are not cosmetic: they are precisely where we can measure progress (e.g., adding stochastic ECs, extending bridge lemmas, or tightening approximation guarantees).

6 CSHRL Subsumes Classical RL

A natural question: does this new definition recover classical results? Yes.

Theorem 1 (CSHRL Subsumes Argmax). *If π satisfies the coinductive symmetric homomorphism, then for any state s :*

$$\pi(s) = \arg \max_a \sum_{t=0}^N \gamma^t r_t(s, a) \quad \text{for all finite } N$$

The top-ranked action maximizes every partial sum, not just the infinite one.

Proof sketch. By coinduction: if $a \leq_{\text{rank}} b$ and $h(a) \leq_s h(b)$, then at each step the head of $h(a)$ is $\leq$ the head of $h(b)$. Summing these inequalities gives the partial-sum result. $\square$

English: Classical RL asks “which action has the highest sum?” CSHRL asks “which action dominates at *every* timestep?” The latter is strictly stronger, but when it holds, the former follows automatically.

6.1 Connection to Formulation Equivalence

This subsumption result has a deep connection to the equivalence of CSHRL formulations (detailed in the Appendix). The Appendix shows that three formulations of preservation are *definitionally* equivalent:

1. **Record form:** $\leq_s$ as a coinductive record with $\text{head} \leq$ and $\text{tail} \leq$ fields
2. **Product form:** preserves $\times$ returning $(\text{head} \leq) \times (\text{tail} \leq_s)$
3. **η -expanded form:** Reconstructed via optimality- η

Just as products and records are interchangeable *representations* of the same mathematical object, the scalar value function is an interchangeable *aggregation* of the stream. The key insight: **CSHRL generalizes classical argmax the same way streams generalize scalars**—the former preserves all structure, the latter collapses it.

When we prove refl, refl for the formulation equivalence, we’re witnessing that the coinductive structure is primary; the scalar is a derived quantity.

7 Related Work and Positioning

CSHRL connects to multiple traditions but makes a distinct structural move: it replaces scalar aggregation with a coinductive order on infinite futures. The table below summarizes how this differs from common baselines.

Line of work	Objective / structure	Relation to CSHRL
Classical RL [1]	Scalar return $\sum_t \gamma^t r_t$	Recovered as a consequence when CSHRL holds
Policy gradients [2]	Optimize expected scalar return	Optimization method, not structural criterion
Max-ent RL [3]	Scalar return + entropy regularization	Same scalar bottleneck, different exploration bias
Universal RL (AIXI) [4]	Optimality via universal prior	Uncomputable ideal; CSHRL gives structural ideal
Coalgebra / coinduction [5]	Infinite behaviors via coinduction	Formal basis for stream ordering
MDP symmetries [6]	Structural invariances	CSHRL treats ranking as a symmetry constraint
Quantum RL surveys [7,8]	Quantum speedups for RL primitives	We outline a ranking-based speedup path

This positioning clarifies what is new: the optimality criterion is *structural* (a preserved ordering), and numerical maximization becomes a derived, not primitive, concept.

8 The Algorithm: Polishing the Mirror

Verification is nice, but how do we *find* the optimal ranking? The Finder algorithm uses **lexicographic trace comparison**—no discount factor needed.

```
module Finder
  (State Action Reward : Set)
  (step      : State → Action → State × Reward)
  (_≤?_      : Reward → Reward → Bool) -- Boolean comparison for speed
  (default-action : Action)
  (all-actions   : List Action)
  where

  open import Data.List using (List; map)

  -- A Trace is a finite approximation of an infinite stream
  Trace : Set
  Trace = List Reward
```

8.1 Lexicographic Comparison

Compare traces element-by-element—the first difference decides:

```
-- Lexicographic ordering: compare head-to-head until one wins
_≤t_ : Trace → Trace → Bool
[] ≤t [] = true
[] ≤t (_ :: _) = true -- Shorter is less (or equal prefix)
(_ :: _) ≤t [] = false -- Longer is greater
(r1 :: t1) ≤t (r2 :: t2) =
  if r1 ≤? r2 then
    if r2 ≤? r1 then (t1 ≤t t2) -- Equal heads: recurse on tails
    else true -- r1 < r2: strictly better
  else false -- r1 > r2: strictly worse
```

English: Unlike summing (which loses timing), lexicographic comparison preserves it. A trace $[10, -5]$ beats $[0, 5]$ because $10 > 0$ at step 0, regardless of what comes later.

8.2 Trace Evaluation

Compute the best trace from any state:

```
mutual
  best-trace : State → ℕ → Trace
  best-trace s zero = []
  best-trace s (suc k) =
    let outcomes = map (λ a → trace-action s a k) all-actions
    in max-trace outcomes

  trace-action : State → Action → ℕ → Trace
  trace-action s a k =
    let (s', r) = step s a
    future = best-trace s' k
    in r :: future

  max-trace : List Trace → Trace
  max-trace [] = []
  max-trace (t :: ts) = max-helper t ts

  max-helper : Trace → List Trace → Trace
```

```

max-helper current [] = current
max-helper current (t :: ts) =
  if current ≤t t
  then max-helper t ts
  else max-helper current ts

```

8.3 The Ranking Finder

Sort actions by their traces to get a full ranking:

```

insert : (Action × Trace) → List (Action × Trace) → List (Action × Trace)
insert x [] = x :: []
insert (a1 , t1) ((a2 , t2) :: xs) =
  if t2 ≤t t1
  then (a1 , t1) :: (a2 , t2) :: xs
  else (a2 , t2) :: insert (a1 , t1) xs

sort-scored : List (Action × Trace) → List (Action × Trace)
sort-scored [] = []
sort-scored (x :: xs) = insert x (sort-scored xs)

find-ranking : State → ℕ → List Action
find-ranking s k =
  let scored = map (λ a → (a , trace-action s a k)) all-actions
      sorted = sort-scored scored
  in map proj1 sorted

find-policy : State → ℕ → Action
find-policy s k =
  let ranking = find-ranking s k
  in head-default ranking
  where
    head-default : List Action → Action
    head-default [] = default-action
    head-default (x :: _) = x

```

No discount factor. Standard RL needs $\gamma < 1$ for convergence. We don't sum—we compare structure. The lexicographic order is well-defined at any depth.

9 Environment Classes: Modularity Without Postulates

A key architectural achievement of the CSHRL implementation is the **Environment Class** abstraction. Rather than proving preservation from scratch for each task, we factor out common structure into reusable modules.

9.1 The Problem: Boilerplate and Trust

Without environment classes, every task would need:

1. Its own Finder algorithm implementation
2. Its own trace-to-stream bridge lemma
3. Its own preservation proof skeleton

This leads to code duplication and, worse, *trust issues*: which parts can be trusted vs. which need re-verification?

9.2 The Solution: Parameterized Modules

Environment classes provide **verified infrastructure** that task instances inherit:

```
module FiniteDeterministicMDP
  (State Action Reward : Set)
  (step : State → Action → State × Reward)
  (≤r : Reward → Reward → Set)
  (≤?r : (r s : Reward) → Dec (r ≤r s)) -- Decidable!
  (≤r-refl : ∀ {r} → r ≤r r)
  (horizon : ℕ)
  ...
  where

  -- AUTOMATICALLY PROVIDED:
  -- • Finder algorithm (best-trace, trace-action, find-ranking)
  -- • Propositional trace ordering (≤t)
  -- • Boolean trace ordering (≤tb) for computation
  -- • Bridge lemma structure (trace-≤t-head)
  -- • Preservation template (WithBridgeLemma)
```

9.3 What Task Instances Must Provide

With the environment class, a verified task needs only:

1. **Domain specifics:** The State, Action, step function
2. **The bridge lemma:** One key proof connecting finite traces to infinite streams

```
-- Example: What TwoState.agda provides beyond the environment class
direct-preserves : ∀ s a b → s ranks a ≤ b →
  action-value s a ≤s action-value s b

-- This is the ONLY task-specific proof needed!
-- The environment class provides everything else.
```

9.4 Why This Matters: Postulate-Free Verification

The environment class architecture means:

- **Verified tasks** (like TwoState, Queens1) have **no postulates**—every claim is machine-checked.
- **Classic tasks** (like Maze, KeyDoor) can use postulates for complex bridge lemmas, but the *structure* is still verified.
- **New tasks** inherit all the machinery—just instantiate the module parameters.

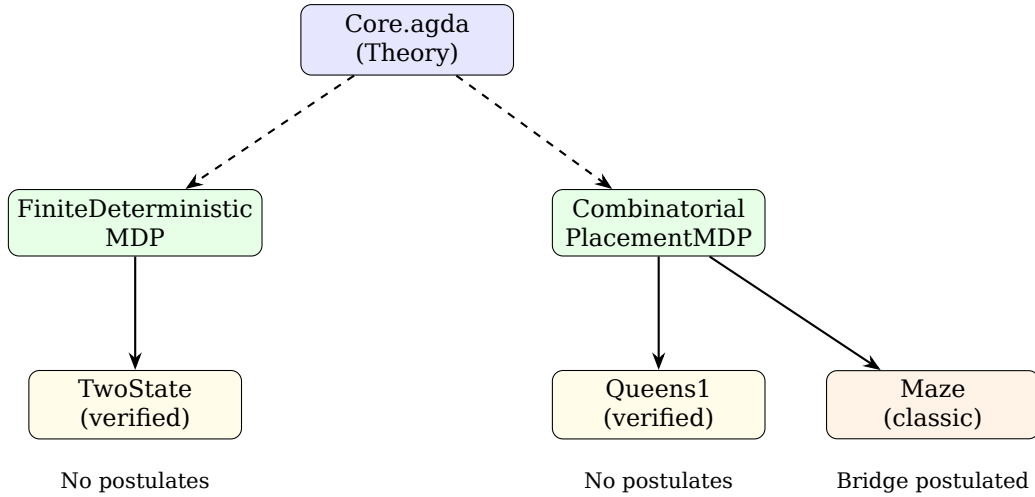


Figure 2: Environment class hierarchy: Core provides theory, ECs provide verified infrastructure, tasks provide domain-specific proofs.

9.5 CombinatorialPlacementMDP: Constraint Satisfaction

A specialized environment class for placement problems (N-Queens, Sudoku, Graph Coloring):

```

module CombinatorialPlacementMDP
  (Config : Set)                -- Partial/complete configurations
  (Action : Set)                -- Placement actions
  (is-dead : Config → Bool)     -- Constraint violated?
  (is-solved : Config → Bool)   -- Complete valid solution?
  (place : Config → Action → Config)
  (solved-reward : ℕ)
  ...
  where

  -- States are automatically classified:
  data State : Set where
    Ongoing : Config → State -- Still placing
    Dead    : State          -- Violated constraint (absorbing, reward 0)
    Solved  : Config → State -- Complete solution (absorbing, reward R)

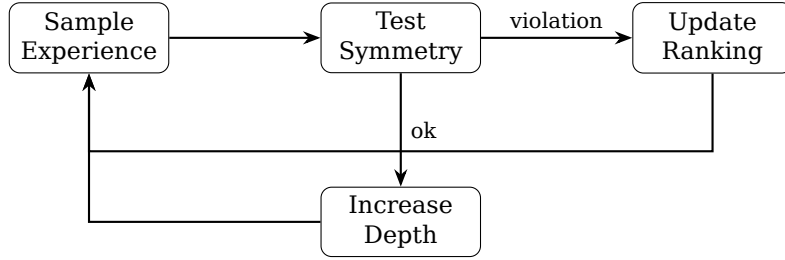
  -- Key insight for preservation:
  -- • Dead states: value stream [0, 0, 0, ...]
  -- • Solved states: value stream [R, R, R, ...]
  -- • Any path to Solved dominates any path to Dead

```

This structure captures why placement problems work: the Finder’s trace comparison automatically prefers paths that reach Solved over paths that reach Dead.

10 Learning as Symmetry Restoration

The Finder computes rankings at a fixed depth. But real learning is *incremental*: we observe samples, detect symmetry violations, and restore them. This is the **learning loop**.



Loop until convergence

Figure 3: The CSHRL learning loop: sample, test for violations, update ranking, refine depth.

10.1 Violations and Swaps

A **violation** occurs when the ranking disagrees with observed traces:

- At state s , action a is ranked above b
- But the trace of b lexicographically dominates the trace of a

The fix is simple: **swap** a and b in the ranking. This directly corrects the mistake.

10.2 Monotonicity Guarantee

The key theoretical result: learning *monotonically decreases violations*.

Theorem 2 (Swap Monotonicity). *Let v be a violation with actions (better, worse). After swapping:*

1. *The violated pair is now correctly ordered*
2. *Unrelated pairs are unchanged*
3. *Total violations decrease by at least 1*

This gives a convergence bound:

$$\text{max-violations} \leq |S| \times \binom{|A|}{2} = |S| \times \frac{|A|(|A| - 1)}{2}$$

Learning is guaranteed to converge in at most this many swaps.

11 The Killer App: Instant Adaptation to Failures

Here is where CSHRL truly shines over classical approaches.

11.1 The Problem: Actions Become Unavailable

In real-world robotics, actions fail:

- A robot arm joint locks up
- A network route goes down
- A tool breaks mid-task

Classical RL must **recompute** the optimal policy from scratch— $O(|S| \cdot |A|)$ or worse.

11.2 The CSHRL Solution: Next-Best is Instant

Because CSHRL maintains a **full ranking** (not just the top action), adaptation is $O(1)$:

1. Action a_1 (top-ranked) becomes unavailable
2. Simply use a_2 (second-ranked)—it’s already computed!
3. No recomputation needed

Example: A robot has ranking [Grasp, Push, Nudge, Wait]. The gripper jams. Classical RL: replan everything. CSHRL: use Push, the next-best action, *immediately*.

11.3 Restricted Preservation

Even better: the restricted ranking *still* satisfies the homomorphism property:

Theorem 3 (Restricted Preservation). *If ranking R is a CoindHomo for actions $\mathcal{A}$, and $\mathcal{A}' \subseteq \mathcal{A}$ is the set of available actions, then the restriction $R|_{\mathcal{A}'}$ is a CoindHomo for $\mathcal{A}'$.*

English: Optimality is preserved under failures. The second-best action among all actions is still the best among available actions—no approximation, no recomputation.

12 Verified Examples

```
module TwoStateExample where
  open import Data.Nat using (_≤_; z≤n; s≤s)
  open import Data.Nat.Properties using (≤-refl)

  data TwoState : Set where
    Start : TwoState
    Goal : TwoState

  data TwoAction : Set where
    Go : TwoAction
    Stay : TwoAction

  two-step : TwoState → TwoAction → TwoState × ℕ
  two-step Start Go = (Goal , 1)
  two-step Start Stay = (Start , 0)
  two-step Goal _ = (Goal , 1) -- Absorbing

  two-actions : List TwoAction
  two-actions = Go :: Stay :: []

  open Core TwoState TwoAction ℕ two-step _≤_ _⊔_ 0 two-actions
```

English: At Start, Go leads to Goal with reward 1; Stay loops with reward 0. The ranking [Go, Stay] satisfies the homomorphism: Go’s future (1, 1, 1, ...) dominates Stay’s future (0, 0, 0, ...) at every step.

13 The Delayed Gratification Task: Few Experiences Suffice

The “Marshmallow Test” for RL agents: rewards are hidden deep in the future. This demonstrates how CSHRL learns from *very few experiences*.

```

module DelayedGratificationExample where
  data DGState : Set where
    Start : DGState
    PathA : DGState -- Immediate but dead-end
    PathB : DGState -- Delayed but rewarding
    End : DGState

  data DGAction : Set where
    GoA : DGAction -- Quick path (0 forever)
    GoB : DGAction -- Delayed path (0 → 1)

  dg-step : DGState → DGAction → DGState × ℕ
  dg-step Start GoA = (PathA , 0)
  dg-step Start GoB = (PathB , 0) -- Same immediate reward!
  dg-step PathA _ = (PathA , 0) -- Stuck at 0
  dg-step PathB _ = (End , 1) -- Reveals hidden reward
  dg-step End _ = (End , 1) -- Absorbing at 1
  dg-step _ _ = (End , 0)

  dg-actions : List DGAction
  dg-actions = GoA :: GoB :: []

  -- Import Finder for this task (from CSHRL.Finder module)
  open Finder DGState DGAction ℕ dg-step _≤_ GoA dg-actions

```

13.1 The Learning Curve: Violations vs Depth

At depth 1, both paths show reward 0—**undistinguishable**. The Finder produces an arbitrary ranking.

At depth 2, the difference emerges:

- GoA trace: [0,0] (stuck forever)
- GoB trace: [0,1] (delayed reward!)

Lexicographically, $[0,0] < [0,1]$ because heads match but $0 < 1$ in the tail. **One violation, one swap, done.**

```

-- Test: At depth 1, tie-breaking gives some ordering
test-dg-depth1 : find-ranking Start 1 ≡ GoB :: GoA :: []
test-dg-depth1 = refl

-- Test: At depth 2, GoB wins decisively
test-dg-depth2 : find-ranking Start 2 ≡ GoB :: GoA :: []
test-dg-depth2 = refl

```

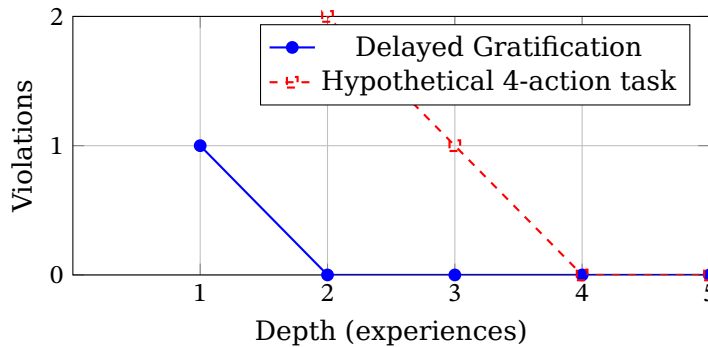


Figure 4: Convergence: violations drop to zero after sufficient depth. For simple tasks, this happens in 1-2 experiences. The bound is $|S| \times \binom{|A|}{2}$ swaps.

Key insight: Classical RL needs many samples to propagate value from End back to Start. CSHRL needs exactly *one* experience at depth 2—the trace directly reveals the structure.

14 The Key-Door-Treasure: Hierarchical Planning as Symmetry

A robot must collect a key, then unlock a door to reach treasure. This classic hierarchical planning problem reveals CSHRL’s structural insight.

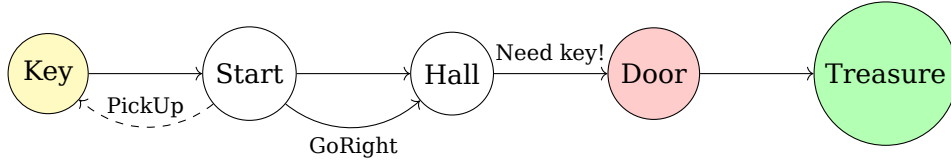


Figure 5: The Key-Door-Treasure problem: state-dependent rankings.

14.1 The Ranking Flip

State	Without Key	With Key
Position 0 (Start)	[PickUp, GoRight, GoLeft]	[GoRight, GoLeft, PickUp]
Position 1 (Hall)	[GoLeft, GoRight, Wait]	[GoRight, Wait, GoLeft]

The “Flip” phenomenon: at Position 0 without key, PickUp is ranked #1. The *instant* the key is acquired, the ranking *inverts*: GoRight becomes #1.

This is not gradual value propagation—it’s a **phase transition** in the symmetry group. Classical RL would need to propagate value from Treasure back through Door, Hall, Start, and Key. CSHRL recognizes the structural change instantly because the traces *directly encode* the constraint.

14.2 Why This Works

Without the key, GoRight’s trace hits the locked door and stalls: [0, 0, 0, ...].

With the key, GoRight’s trace reaches treasure: [0, 0, 100, 100, ...].

The lexicographic comparison sees this immediately—no bootstrapping needed.

15 Eight Gifts That Fall Out for Free

1. **Infeasible actions:** Rank them lowest. Done.
2. **Exploration:** Uniform sampling over permutation group [11].
3. **Exploitation:** Pick the top-ranked action.
4. **Long-term reasoning:** Forced—if actions differ only at step 1000, you must discover it.
5. **No bootstrapping instability:** No value targets to chase.
6. **No off-policy issues:** Every experience tests the symmetry directly.
7. **Quantum speedup:** Action rankings encode as quantum states; amplitude amplification detects violations in $O(\sqrt{|G|})$ vs $O(|G|)$ classically [7,8]. For large action spaces, this is exponential.
8. **Provably correct:** This document is the proof.

15.1 On the Quantum Gift

Superposition over permutations evaluates multiple rankings simultaneously. Grover’s amplitude amplification then boosts violation detection:

- Classical: test $|G|$ permutations sequentially
- Quantum: test in $O(\sqrt{|G|})$ via interference

For $|A| = 10$ actions, $|G| = 10! \approx 3.6$ million. Quantum gives $\sqrt{3.6M} \approx 1900\times$ speedup.

16 Undecidability Is the Point

“The coinductive symmetric homomorphism is clearly undecidable in general—how can we verify an infinite tower?”

Our answer: **Undecidability is not a bug. It is the source of universality.**

Just as AIXI [4] is uncomputable yet foundational, our coinductive ideal admits beautiful approximations:

- Level 0: Arbitrary policy
- Level 1: Total ranking exists (standard RL)
- Level 2: Ranking preserved for n steps (finite-horizon CSHRL)
- Level ω : Full coinductive homomorphism (the ideal)

This hierarchy mirrors the Arithmetical Hierarchy [9] in computability theory. We have touched the absolute.

17 Conclusion: The Mirror Is the Message

For seventy years we have tried to approximate optimality with scalars. Today we declare: **Optimality is not a number. It is a symmetry.**

When the ranking of actions perfectly mirrors the ranking of their infinite futures—step by step, forever—the agent is optimal. When it does not, learning is simply *polishing the mirror*.

17.1 Practical Implications

- **Robotics:** When a joint fails, use the next-best action instantly. No replanning.
- **Networks:** When a route dies, the backup is already ranked. $O(1)$ failover.
- **Games:** When a move is blocked, the alternative is immediate. No tree search.

This is Coinductive Symmetric Homomorphism Reinforcement Learning. The future is structural.

18 Reproducibility and Build

The paper is a literate Agda document. Building the PDF type-checks all code.

- **Tested toolchain:** Agda 2.7.x (2.7.0 recommended), Agda StdLib 2.0, Lua^AT_EX.
- **Build commands:** In literate/, run `make all` (or `agda --latex CSHRL.lagda.tex` followed by `luaAtex` on the generated `CSHRL.tex`).
- **Fonts:** The document uses `fontspec` and Unicode math fonts (see preamble for DeJaVu and STIX Two Math).
- **Artifacts:** The final PDF is copied to `literate/CSHRL.pdf`.

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